```python
import numpy as np

# Load the dataset
# Assuming 'Sample.csv' has columns: RollNo, Subject1, Subject2, Subject3
try:
    data = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
except FileNotFoundError:
    print("Error: Sample.csv not found. Please ensure the file is in the same
directory.")
    # Creating dummy data for demonstration if file is missing
    data = np.array([
        [301, 67, 77, 88], [302, 78, 88, 77], [303, 45, 56, 89],
        [304, 88, 98, 45], [305, 78, 88, 99], [306, 68, 78, 36],
        [307, 79, 12, 45], [308, 46, 45, 77], [309, 89, 32, 88],
        [310, 79, 24, 33]
    ])

# 1. Print all student details
print("All Student Details:\n", data)

# 2. Find total students
print("\nTotal Students:", data.shape[0])

# 3. Print all student roll numbers
print("All Student Roll Nos:", data[:, 0])

# 4. Print Subject 1 marks
print("Subject 1 Marks:", data[:, 1])

# 5. Find minimum marks in Subject 2
print("Min marks in Subject 2:", np.min(data[:, 2]))

# 6. Find maximum marks in Subject 3
print("Max marks in Subject 3:", np.max(data[:, 3]))

# 7. Print all subject marks (excluding roll numbers)
print("All subject marks:\n", data[:, 1:])

# 8. Find total marks of students (row-wise sum)
total_marks = np.sum(data[:, 1:], axis=1)
print("Total Marks per student:", total_marks)

# 9. Find average marks of each student
avg_per_student = np.mean(data[:, 1:], axis=1)
print("Average Marks per student:", np.round(avg_per_student, 2))

# 10. Find average marks of each subject (column-wise mean)
avg_per_subject = np.mean(data[:, 1:], axis=0)
print("Average Marks of each subject:", avg_per_subject)

# 11. Find average marks of Subject 1 and Subject 2
print("Average Marks of S1 and S2:", np.mean(data[:, 1:3], axis=0))

# 12. Find average marks of Subject 1 and Subject 3
```

```
print("Average Marks of S1 and S3:", np.mean(data[:, [1, 3]], axis=0))

# 13. Roll number of student with maximum marks in Subject 3
max_s3_idx = np.argmax(data[:, 3])
print("Roll No with Max Marks in S3:", data[max_s3_idx, 0])

# 14. Roll number of student with minimum marks in Subject 2
min_s2_idx = np.argmin(data[:, 2])
print("Roll No with Min Marks in S2:", data[min_s2_idx, 0])

# 15. Roll number of students who scored exactly 24 in Subject 2
s2_24_mask = data[:, 2] == 24
print("Roll No(s) who scored 24 in S2:", data[s2_24_mask, 0])

# 16. Count students with < 40 in Subject 1
count_less_40_s1 = np.sum(data[:, 1] < 40)
print("Count < 40 in S1:", count_less_40_s1)

# 17. Count students with > 90 in Subject 2
count_more_90_s2 = np.sum(data[:, 2] > 90)
print("Count > 90 in S2:", count_more_90_s2)

# 18. Count students who scored >= 90 in each subject
# (Checks if all marks in a row are >= 90)
all_above_90 = np.sum(np.all(data[:, 1:] >= 90, axis=1))
print("Students with >= 90 in ALL subjects:", all_above_90)

# 19. Count of subjects in which each student scored >= 90
subjects_above_90 = np.sum(data[:, 1:] >= 90, axis=0)
print("Count of subjects where every student scored >= 90:",
np.sum(subjects_above_90 == data.shape[0]))

# 20. Print Subject 1 marks in ascending order
sorted_s1 = np.sort(data[:, 1])
print("Subject 1 marks in ascending order:", sorted_s1)
```

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.

= RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python314/assigment 1.py
Error: Sample.csv not found. Please ensure the file is in the same directory.
All Student Details:
 [[301  67  77  88]
 [302  78  88  77]
 [303  45  56  89]
 [304  88  98  45]
 [305  78  88  99]
 [306  68  78  36]
 [307  79  12  45]
 [308  46  45  77]
 [309  89  32  88]
 [310  79  24  33]]

Total Students: 10
All Student Roll Nos: [301 302 303 304 305 306 307 308 309 310]
Subject 1 Marks: [67 78 45 88 78 68 79 46 89 79]
Min marks in Subject 2: 12
Max marks in Subject 3: 99
All subject marks:
 [[67 77 88]
 [78 88 77]
 [45 56 89]
 [88 98 45]
 [78 88 99]
 [68 78 36]
 [79 12 45]
 [46 45 77]
 [89 32 88]
 [79 24 33]]
Total Marks per student: [232 243 190 231 265 182 136 168 209 136]
Average Marks per student: [77.33 81.   63.33 77.   88.33 60.67 45.33 56.
69.67 45.33]
Average Marks of each subject: [71.7 59.8 67.7]
Average Marks of S1 and S2: [71.7 59.8]
Average Marks of S1 and S3: [71.7 67.7]
Roll No with Max Marks in S3: 305
Roll No with Min Marks in S2: 307
Roll No(s) who scored 24 in S2: [310]
Count < 40 in S1: 0
Count > 90 in S2: 1
Students with >= 90 in ALL subjects: 0
Count of subjects where every student scored >= 90: 0
Subject 1 marks in ascending order: [45 46 67 68 78 78 79 79 88 89]
```